

Le Hoang Tran

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SUMMARY

Backend-focused Software Engineer with strong experience in building large-scale systems. Specialized in high-concurrency, low-latency, cloud-native infrastructure and performance optimization. Delivered products that serve millions of users.

EXPERIENCE

Viettel Group

May 2021 – Jun 2025

Backend Software Engineer

Hanoi, Vietnam

Customer Engagement Platform

2023 – 2025

- Designed and implemented an in-house **marketing system** to support personalized campaigns via push notifications, in-app messaging, and rewards distribution, delivering **100M+ events** to end users daily.
- Optimized user segmentation creation throughput from **500 to 35,000 QPS**, **reducing p95 latency by 80%** by introducing **Roaring Bitmap**-based solution and multiple caching layers.
- Engineered **workflow automation module** using Camunda, including business process orchestration, rule evaluation, timer events, enabling non-technical operations teams to configure marketing workflows independently.
- Led a team of 4 backend engineers**; driving sprint planning, system design and code reviews to continuously extend system functionality, in response to user needs and growing demands.
- Monitored system metrics via Grafana** and proactively detected potential areas for performance improvement. Reported to relevant teams, applied multiple **optimization techniques** including configuration tuning, caching strategies, and microservices patterns.

Data Tracking System

2022 – 2023

- Designed and delivered a high-throughput in-house **event tracking system** for mobile and web platforms using Java and Spring Boot, processing **150M+ events/day** and **eliminating third-party licensing costs**.
- Implemented an online tracking rule configuration module, supporting **3,000+ dynamic tracking rules**.
- Reduced data-capture latency by 30%** by introducing decision tree-based engine to optimize validation process
- Engineered reactive data-fetching APIs with **Spring WebFlux** to support high-concurrency workloads, handling over **150M+ events daily**.

Smart Authentication System

2021 – 2022

- Collaborated on the research and delivery of **SmartOTP** solution for **20M+ end users**, eliminating reliance on SMS OTPs, **saving over \$200K annually** and enhancing authentication reliability.
- Implemented secure key exchange, OTP generation, and digital signature mechanisms across critical APIs, strengthening authentication and API security.
- Applied **mobile security hardening techniques** (code obfuscation, root/jailbreak/tamper detection) to Android application using native **C/JNI**, passing enterprise-level security audits.

User Contact System

2024

- Designed and implemented a backend module to collect and synchronize user contact data across multiple devices for the Viettel Money application, supporting **20M+ end users**.
- Optimized contact storage with Roaring Bitmap, reducing storage usage by **70%** and lowering user matching latency

EDUCATION

George Mason University

(Expected) Dec 2026

M.S. in Computer Science | GPA: 4.0

Fairfax, VA, USA

Hanoi University of Science and Technology

May 2017 – May 2021

B.S. in Information Technology | GPA: top 5% of major

Hanoi, Vietnam

SKILLS

Languages: Java, Go, C, JavaScript, Python

Backend: Spring Framework, Apache Kafka, Camunda, REST, gRPC, Unit & Integration Testing

Databases: MySQL, MongoDB, Redis/Valkey, Elasticsearch, MinIO, Clickhouse

Infrastructure: AWS, Docker, Kubernetes, CI/CD, Jenkins, SonarQube, Keycloak, ELK, k6

Familiar with: Agile SDLC, Distributed Systems, Design patterns, Cybersecurity, AI-assisted development

CERTIFICATES

AWS Certified Solutions Architect – Associate

Sep 2023

Oracle Certified Java SE Programmer

Apr 2023